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Clinical evaluation of a wrist sensorimotor rehabilitation program for triangular fibrocartilage complex injuries

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Abstract

Introduction: Triangular fibrocartilage complex (TFCC) injuries are associated with distal radioulnar joint (DRUJ) instability and impaired wrist proprioception. Sensorimotor training of extensor carpi ulnaris (ECU) and pronator quadratus (PQ) can enhance DRUJ stability. With limited evidence on effectiveness of TFCC sensorimotor rehabilitation, this study aimed to evaluate the effects and feasibility of a novel wrist sensorimotor rehabilitation program (WSRP) for TFCC injuries.

Methods: Patients diagnosed with TFCC injuries were recruited from May 2018 to January 2020 at an outpatient hand clinic in Singapore General Hospital. There are four stages in WSRP: (1) pain control, (2) muscle re-education and joint awareness, (3) neuromuscular rehabilitation, and (4) movement normalization and function. WSRP also incorporated dart throwing motion and proprioceptive neuromuscular facilitation. Outcome measures included grip strength measured with grip dynamometer, numerical pain rating scale, joint position sense (JPS) measurement, weight bearing measured with the 'push-off' test, and wrist function reported on the Patient Rated Wrist Hand Evaluation.

Results: Ten patients completed the WSRP. Mean changes were compared with minimal clinically important differences (MCID) for outcomes. All patients achieved MCID on pain, 70% of patients achieved MCID on grip strength, weight bearing and wrist function. Paired t-tests and Cohen's D for outcome measures were calculated. There were large effect sizes of 2.47, 1.35, and 2.81 for function, grip strength and pain respectively, and moderate effect sizes of 0.72 and 0.39 for axial loading and JPS respectively.

Discussion: WSRP presents a potential treatment approach in TFCC rehabilitation. There is a need for future prospective clinical trials with control groups.

Keywords

Sensorimotor rehabilitation, triangular fibrocartilage complex, wrist, proprioception

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Introduction

The triangular fibrocartilage complex (TFCC) comprises the dorsal and palmar radioulnar ligament, central articular disc, the ulnocarpal ligaments, the extensor carpi ulnaris tendon subsheath, and the meniscal homolog.[1] TFCC stabilizes the ulnar side of radiocarpal and the distal radioulnar joint (DRUJ).[2,3] As DRUJ is a load-bearing joint, TFCC also has an important role to cushion compressive loads transmitted from the ulnar carpus, and assist with load transmission between the carpus and ulna.[4,5] Studies have reported that injury to the TFCC results in at least 20% reduction in the ability to bear load.[3] The TFCC is also found to be richly innervated by nerves which transmit reflexes received from mechanoreceptors present at the wrist joint.[6–9] Injury to TFCC has been associated with decreased wrist proprioception.[10] Patients with TFCC injury may experience reduced

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wrist sensorimotor control due to decreased wrist proprioception, in addition to DRUJ instability which causes ulnar sided wrist pain, decreased grip strength and reduced ability to weight bear.[2,10,11] These lead to problems with wrist function in daily activities.

Kinesthesia, joint position sense (JPS), and neuromuscular control make up the human sensorimotor system for proprioception.[8] Kinesthesia and JPS bring conscious awareness of joint positions with inputs from mechanoreceptors present in the wrist joint.[9] Neuromuscular control allows for subconscious control of the joint with reflex muscle responses at spinal cord and subsequent muscle regulation at the cerebrum.[8] A study on TFCC injuries recommended neuromuscular training of DRUJ stabilizers extensor carpi ulnaris (ECU) and pronator quadratus (PQ).[12] Researchers have found that ECU and PQ constrained volar and dorsal displacement of the DRUJ during loading.[1] Contraction of PQ coapts the distal ulna into the sigmoid notch of the distal radius and shifts the ulna proximally to prevent the head of ulna from impacting against the carpal bone during loading which could otherwise cause pain, while ECU resists abnormal displacement of ulnar head.[13,14] With adequate sensorimotor control of ECU and PQ, DRUJ can be stabilized to allow pain-free participation in daily activities.[1,3,13,15]

At other joints such as the ankle or shoulder, sensorimotor intervention is an established component in rehabilitation programs for joint stability.[16,17] Sensorimotor rehabilitation for wrist ligament injuries has only garnered interest in recent years. In the literature, there are descriptions of sensorimotor rehabilitation techniques involving neuromuscular rehabilitation of specific joint-stabilizing muscles to re-establish wrist stability.[12,18] However, there is a dearth of evidence on the efficacy of sensorimotor rehabilitation programs. In recent years, a service evaluation on the Birmingham Wrist Instability Programme targeting scapholunate ligament injury found clinically important changes in pain, grip strength and function.[19] Similarly, a pilot study investigating the distal radioulnar stability training (DRUST) program for TFCC injuries showed promising effects in reducing pain and increasing functions.[20] In both studies, sensorimotor rehabilitation occurred in stages with a gradual progression of training difficulty.[19,20]

Wrist sensorimotor rehabilitation program (WSRP)

This study describes a novel wrist sensorimotor rehabilitation program (WSRP) for patients with TFCC injury. WSRP is a pain-directed program with clear criteria for gradual progression to prevent injury. WSRP was developed based on sensorimotor control principles described in the literature. Researchers described two main phases in sensorimotor rehabilitation for wrist ligament injuries.[8,12] The early phase focuses on pain control with protected exercises, while the late phase works on balance and reactive exercises when adequate tissue healing had occurred.[8,12] WSRP focuses on guided transition from early phase of sensorimotor rehabilitation with controlled exercises at the wrist to late phase of sensorimotor rehabilitation with exercises targeting the entire upper extremity. Dosage of exercises in WSRP is in line with recommendations described in the literature for initial training of muscular hypertrophy and endurance after an injury.[21] There are four stages in WSRP: (1) pain control, (2) muscle re-education and joint awareness, (3) neuromuscular rehabilitation, and (4) movement normalization and function (Table 1). The transitions progressively prepare patients to return to their premorbid level of activities with their injured hand. Rehabilitation techniques at the early stages involve isometric activation of PQ and ECU, and dart throwing motion (DTM) mobilization and strengthening exercises. To isolate PQ, patients are instructed to pronate the forearm with their contralateral hand positioned at the distal radius to resist pronation (Figure 1(a)). For ECU isometric contraction, patients are instructed to ulnarly extend the wrist against the contralateral fingers positioned on the dorsum of 4th and 5th metacarpal shafts which provide a counter force to the ulnar wrist movement (Figure 1 (b)). Neuromuscular rehabilitation of DRUJ stabilizers PQ and ECU and reactive muscle activations exercises enhance DRUJ stability by re-establishing proprioceptive reflexes at the joint to react to destabilizing forces in daily activities.[2] In WSRP, DTM exercises are introduced early to help restore wrist movements. Biomechanical studies have found that the mechanical axis of wrist and the orientation of muscle insertions is oriented along the DTM plane.[22–24] This finding suggests that wrist motion at the DTM plane can be performed with the least effort. Moreover, with the wrist moving predominantly in the oblique plane of the DTM during daily activities,[25,26] DTM exercise begins to prepare the wrist for function. At the later stages in WSRP, rehabilitation techniques involve proprioceptive neuromuscular facilitation (PNF) and specific movement training based on patients' needs. The use of PNF techniques trains multi-joint coordination of the upper extremity in functional movements patterns. The PNF diagonal 2 flexion-extension pattern of the upper extremity also integrates isolated DTM exercise at the wrist into the kinetic chain of the upper extremity.[27] In the clinic, the use of the Baltimore Therapeutic Equipment (BTE) further trains patient-identified movement patterns in their

Table 1. Rehabilitation techniques in Wrist Sensorimotor Rehabilitation Program (WSRP)

Stage of intervention	Rehabilitation goals	Rehabilitation techniques	Exercise examples	Frequency	Criteria for progression
Stage 1	Pain control	DTM AROM Interval splinting		20 reps x 3-5 sets daily	No pain with full AROM NPRS < 2 with isometric
Stage 2	Muscle re-education, joint awareness	Continue DTM AROM Isometric ECU & PQ Isotonic DTM Night splinting	Self-applied, forearm in neutral (Figure 1) With 0.5kg weighted ball (Figure 2)	20 reps x 3-5 sets daily 30 s hold x 5 sets daily 10-20 reps x 3 sets 3-4x/wk	NPRS <2 with isotonic Able to complete 10 reps x 2 sets of exercise Able to maintain control through range
Stage 3	Neuromuscular rehabilitation	Continue isometric ECU & PQ, progress to isotonic Reactive muscle activations Graded weight bearing Coordination training Off splint	 Weighted ball flip Flexbar displacement (Figure 3) Loading on gym ball against wall, +/- perturbations PNF with free weights	10-20 reps x 5 sets 3-4x/wk 2-3 mins x 3 sets 3-4x/wk 10-20 reps x 3 sets 3-4x/wk 10-20 reps x 3 sets 3-4x/wk	NPRS <2 with weight bearing (wall) Good multi-joint coordination
Stage 4	Movement normalization & functions	Stop isometric & isotonic exercises Continue reactive muscle activations Progress weight bearing Resistive coordination training	 Weighted slosh tube Power ball Wall push up Ground push up with push-up bar PNF with theraband (Figure 4) BTE simulations in clinic	 2-3 mins x 3 sets 3-4x/wk 10-20 reps x 3 sets 3-4x/wk 10-20 reps x 3 sets 3-4x/wk	Return to pre-morbid level of upper extremity functions

AROM= active range of motion; BTE= Baltimore Therapeutic Equipment; DTM= dart throwing motion; ECU= extensor carpi ulnaris; PNF= proprioceptive neuromuscular facilitation; PQ= pronator quadratus; NPRS = numerical pain rating scale; wk: week

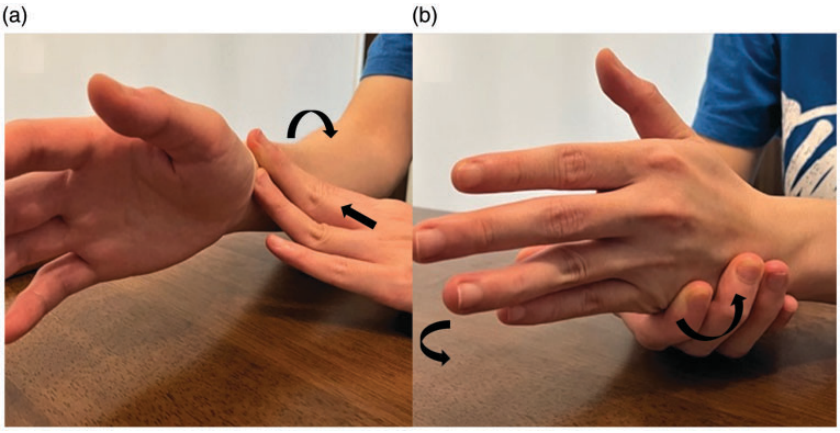


Figure 1. Isometric PQ and ECU, with arrows illustrating direction of forces. (a) Isometric PQ; (b) Isometric ECU.

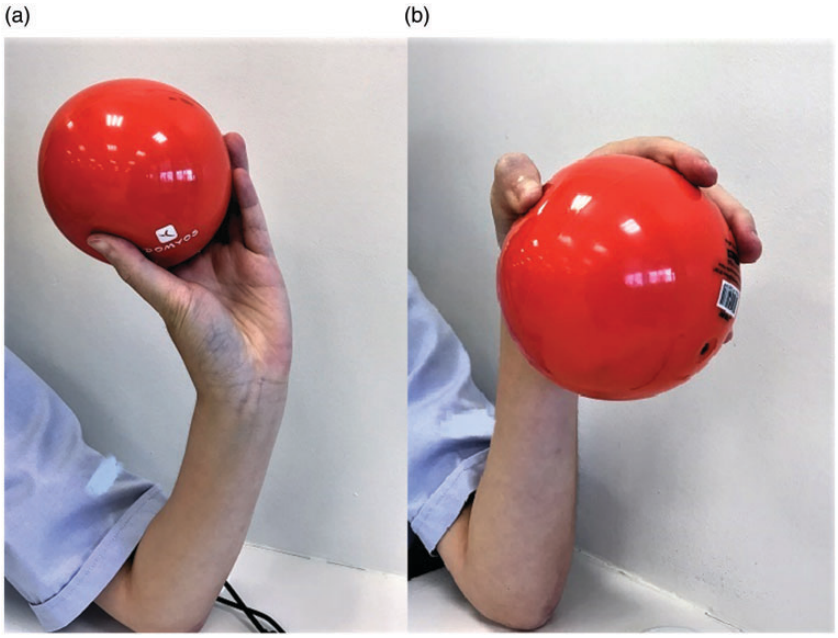


Figure 2. Isotonic DTM. (a) Start position; (b) end position.

activities for movement efficiency.[28] Researchers have recommended movement-specific strengthening to condition the extremity for efficiency in daily life.[29]

Purpose of study

The primary objective was to evaluate the effects of a novel WSRP in reducing pain and improving wrist function, weight bearing, grip strength and JPS in a prospective study on patients with TFCC injuries. We adapted and refined the TFCC wrist sensorimotor rehabilitation program first described by Chen in a retrospective case report.[30] The secondary objective was to test the feasibility of implementing WSRP. We hypothesized that at least 50% of patients who participated in WSRP would achieve minimal clinically important difference (MCIDs) in all outcome measures used.

Methods

Study design and participants

A single group pre-post experimental design was used. Patients were recruited at an outpatient clinic located in Singapore General Hospital between May 2018 to January 2020. Patients were diagnosed with TFCC injury after clinical examination and/or magnetic resonance imaging by their hand surgeons in clinic and were referred to outpatient hand therapy clinic for rehabilitation without operative intervention. Clinical



Figure 3. Reactive muscle activations-flexbar displacement.

examination comprised testing tenderness at the fovea and assessing the stability of the DRUJ. Tenderness at the fovea denotes a positive ulnar fovea sign.[31] The positive ulnar fovea sign was reported to have a 95% sensitivity and 86% specificity in clinical detection of fovea disruptions or ulnotriquetral ligament tears.[31]

Inclusion criteria included patients aged between 21–65 years old who had sustained a TFCC injury within the last one year. Exclusion criteria were TFCC injury with concomitant wrist fractures and/or other wrist ligament injuries. Written informed consent was obtained from each patient prior commencement of intervention, and the rights of patients were protected. Ethics approval was obtained from Singhealth Institutional Review Board (reference number: 2017/3149).

Intervention

The pain-directed treatment approach in WSRP enabled patients to take control of their rehabilitation process with prescribed home exercises. Exercise progression was guided by the clinicians' clinical reasoning, the patients' symptom response to prescribed exercises, and the patients meeting defined criteria established at each stage. Each session began with a careful reevaluation of the patient's wrist. Patients were asked to demonstrate their home exercises to

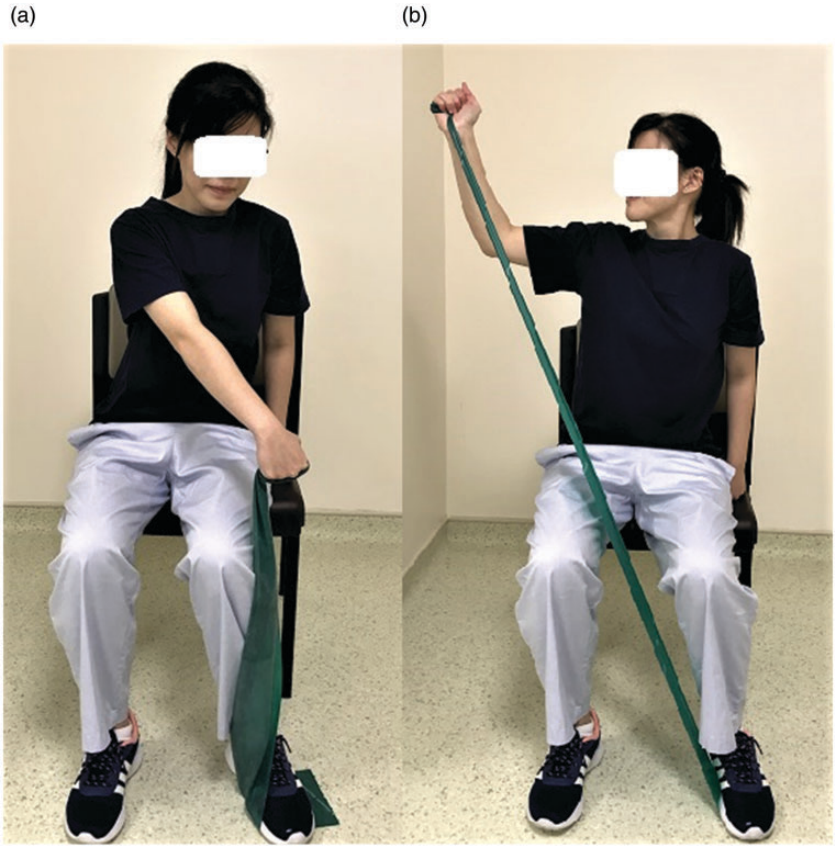


Figure 4. Resistive coordination training- PNF with theraband. (a) start position; (b) end position.

assess understanding of the exercises. If patients did not perform the home exercises accurately, their exercises were not progressed at that session. During each session, patients performed all exercises with direct supervision to ensure competence to perform at home. During the exercise, wrist pain was monitored closely and was not allowed to exceed 2 points on the Numerical Pain Rating Scale (NPRS) to ensure relatively pain-free training and prevent exacerbation of symptoms with training.[21] Quality of movement was emphasized, and exercises only advanced if patients were able to demonstrate smooth coordinated movements without compensations proximally.

The length of time between sessions varied and was decided between the therapist and the patient. The total duration of WSRP varied for each patient dependent on the patient's symptoms and goals. For patients with DRUJ instability and who had to be immobilized in a below elbow Muenster orthosis for 4–6 weeks as recommended by their hand surgeons, WSRP only commenced after the period of full-time splinting.

Outcome measures

Pain during motion. The wrist was examined in a systematic sequence: active range of motion, passive overpressure, and isometric resisted range of motion in cardinal planes of wrist and forearm movements. This systematic sequence of examination has been suggested by Porretto-Loehrke et al. for a comprehensive evaluation of the wrist joint.[32] Wrist pain during these maneuvers was measured with NPRS on a continuum of a 11-point (0–10) scale, with 0 being no pain and 10 being most pain. In the upper extremity the MCID for NPRS has been established as 2 points.[33]

Grip strength. We measured grip strength using a JAMAR dynamometer (Sammons Preston Rolyan, Chicago, IL, USA) according to published standards with the handle of the dynamometer set on the second position.[34] To obtain the highest test-retest reliability suggested in the literature, the average of 3 attempts of maximal grip strength measured on each hand was obtained.[34] The MCID for grip strength in patients with wrist injuries was reported to be 6.5 kg.[35]

Proprioception. Joint position sense (JPS) measures a joint's ability to reproduce a specific joint angle.[12] Karagiannopoulos et al. has described the procedures to measure JPS in his study.[4] In this study, we adapted his technique. The patient positioned their elbow on a table with the forearm in neutral and vision occluded. The therapist passively moved the affected wrist to specified angles of 20 degrees and 30 degrees of wrist extension and then placed it back in neutral. The patients were asked to actively reproduce the exact joint angles on the same wrist respectively. Each joint angle was measured once using a wrist goniometer (Smith & Nephew srl, Agrate Brianza, MI, Italy), and the absolute difference between the positioned angle and reproduced angle was determined. The average of two readings was obtained. In JPS assessment at the wrist, an improvement of at least 7 degrees is considered clinically important.[4]

Weight-bearing. The ability to weight bear through the wrist was measured using a JAMAR dynamometer according to the setup and measurement described in the Push-off test developed by Vincent et al.[5] The handle of the dynamometer was set on the second position and reversed with the convex side facing upwards. The dynamometer was stabilized on a stable surface, and the patient pushed down onto the dynamometer handle with the test arm positioned in about 20 degrees shoulder extension and elbow flexion respectively, and the wrist and forearm in comfortable positions as determined by the patient. An average of three readings was obtained. Minimal detectable change (MDC) for the Push-off test was established to be 4.4 kg.[36]

Wrist function. The functional capability of the wrist in performance of daily activities was assessed with the Patient Rated Wrist Hand Evaluation (PRWHE). The 15-item PRWHE questionnaire has demonstrated good reliability, validity, and internal consistency, and is able to reflect limitations in wrist functions.[37] The PRWHE has an established MCID value of 14 points in the wrist population.[38]

Data collection

Demographic data such as age, sex, time since injury were collected from medical records and/or interview with the patients during their first visit. Outcome measures were recorded by the treating therapist at baseline and on discharge. Duration between initial assessment and discharge was recorded to track total number of days to complete WSRP.

Data analysis

All analyses were performed using Microsoft Excel 2002 Analysis ToolPak. Patients' demographics were summarized with mean and standard deviation (SD) or numbers and percentage where appropriate. Median was used to calculate total duration to complete WSRP as one patient's participation in WSRP was affected by COVID-19 lockdown, resulting in a large outlier data. Absolute change in score for each patient in each outcome measure was compared with

outcome MCID or MDC. The percentage of patients who achieved each outcome MCIDs was calculated. We also performed subgroup analysis for patients with 4–6 weeks of splinting and patients with no splinting. The percentage of patients who achieved outcome MCIDs in each group was calculated. In addition, paired *t*-tests and Cohen's D were calculated for each outcome measure. In this study, an effect size greater than 0.80 indicates a large clinical effect, 0.50 indicates a moderate clinical effect, and 0.20 indicates a weak effect.[39] Where a *P* value was calculated, the level of significance was set at $P \leq .05$.

Results

A total of 13 patients were recruited, and 10 patients completed the WSRP. Two patients were lost to follow-up and one patient opted for operative management following discussion with the hand surgeon. Of the 10 patients who completed WSRP, three patients had to complete 4–6 weeks of full time splinting prior to commencement of WSRP. The mean age of patients was 33.5 years old (SD 9.5), 60% of the patients were female, while 40% were male. The mean time since injury was 5.3 months (SD 3.4). The median duration to complete WSRP was 109 days (IQR 103). There were no reported adverse events during participation in WSRP.

Outcomes

Seventy percent of patients demonstrated clinical improvement greater than the MCID in PRWHE, grip strength and weight bearing. For pain, 100% patients had improvement greater than the MCID (Table 2). Our statistical analysis yielded large effect sizes of 2.47, 1.35, and 2.81 for function, grip strength and pain respectively, and moderate effect sizes of 0.72 and 0.39 for weight bearing and JPS respectively (Table 2). The percentage of patients achieving outcome MCIDs in patients who had 4–6 weeks of splinting and those who did not undergo splinting are presented in Table 3.

Discussion

This study demonstrated the positive effects of WSRP on pain, grip strength, ability to weight bear, and wrist function. In this study, 100% of patients achieved the MCID for pain. There was a large effect size of 2.8. Arguably, WSRP's true effect on pain cannot be determined in this study without a comparative group. However, Bonhof-Jansen et al. suggested in their study that natural healing was unlikely to reduce pain symptoms in patients who sustained the injury more than 3 months ago.[20] In their study, 13 patients who sustained the injury more than 3 months ago made clinical improvements in pain after the DRUST

Table 2. Pre and post treatment mean outcomes.

Outcome	Pre treatment mean (SD)	Post treatment mean (SD)	Absolute mean change	Cohen's D	P value	Published MCID/MDC	% pts achieved MCID
PRWHE, points	35.70 (14.07)	10.45 (6.34)	25.25	2.47	<0.001	14[38]	70
Grip strength, kg	17.80 (8.01)	27.33 (6.15)	9.53	1.35	<0.001	6.5[35]	70
Pain, points	2.50 (1.78)	0.00 (0)	2.50	2.81	<0.01	2[33]	100
Weight-bearing, kg	15.23 (9.45)	22.10 (9.54)	6.87	0.72	<0.01	4.4[36]	70
JPS, degrees	6.20 (2.74)	5.20 (2.44)	1.00	0.39	0.45	7[4]	10

%: percentage; JPS: joint position sense; MCID: minimal clinically important difference; MDC: minimal detectable change; PRWHE: Patient Rated Wrist Hand Evaluation; pts: patients.

Table 3. Percentage of patients achieving outcome MCIDs in patients with splinting and patients without splinting.

		Patients with splinting		Patients without splinting	
Outcome	Published MCID/MDC	Mean change	% pts achieved MCID	Mean change	% pts achieved MCID
PRWHE, points	14[38]	33.83	100	21.57	57.14
Grip strength, kg	6.5[35]	9.33	100	9.62	57.14
Pain, points	2[33]	4.67	100	2.14	100
Weight bearing, kg	4.4[36]	6.33	66.67	7.10	71.43
JPS, degrees	7[4]	4.00	33.33	0.29	14.29

%: percentage; JPS: joint position sense; pts: patients; MCID: minimal clinically important difference; MDC: minimal detectable change; PRWHE: Patient Rated Wrist Hand Evaluation.

program.[20] In this study, the majority of patients sustained the injury more than 3 months ago, with a mean of 5.3 months.

Additionally, all patients in this study did not report any aggravation of pain during their participation in WSRP. The WSRP pain-directed approach with wrist pain not exceeding two points on NPRS during exercise could have ensured that patients were able to cope with the prescribed exercise intensity without overexerting. Careful evaluation of patient's accuracy in performing home exercises at every session ensured that specific DRUJ-stabilizing muscles PQ and ECU were trained in a sequential manner to prevent injury. The defined criteria at each stage for exercise progression could also have prevented training at a level too advanced for patients that might have resulted in maladaptive movement patterns due to inadequate sensorimotor control which could lead to pain exacerbation.[40]

Researchers have suggested that pain has an inverse relationship with grip strength.[41] With improvement in pain, grip strength should improve. In this study, 70% of patients achieved clinical improvement in grip strength, and there was a large effect size of 1.35. Researchers have found that isolated training at the joint can strengthen the weaker muscles but training of the entire extremity is necessary to build up muscular performance in daily activities.[42] This concurs with the WSRP which emphasizes the transition from isolated wrist exercises to training of the entire upper extremity.

With the DRUJ being a load bearing mechanism, the patient's ability to weight bear is pertinent for function. In this study, 70% of patients achieved clinical improvement in weight bearing ability, with a moderate effect size of 0.72. Researchers have hypothesized that during weight bearing across the wrist, proper activation of PQ and ECU may absorb the load at the DRUJ to create a force equilibrium at the joint for stability to weight bear.[3,15] Neuromuscular rehabilitation of DRUJ stabilizers PQ and ECU could possibly have improved the wrist's ability to bear weight.

We measured JPS in one axis of wrist flexion-extension angle. Park and colleagues demonstrated in their study that only proprioception of forearm rotation was affected in TFCC injury.[10] They found that there was no significant difference in JPS of wrist flexion-extension and radioulnar deviation between wrists with TFCC injury and normal wrists. This could explain our non-significant results in JPS with only 10% patients achieving MCID.

For JPS outcome, we used the MCID established for the wrist fracture population.[4] This is the only MCID established for JPS to date. Using MCID derived from the wrist fracture populations may not be an appropriate criterion for patients with TFCC injury given the difference in nature of the injury. Pain has been reported to be a significant contributor to JPS deficits.[4] One could presume the type and amount of pain experienced for patients who sustained a wrist fracture compared to a ligament injury would differ. The baseline for JPS deficit and the degree of deficit change with intervention would correspondingly differ for these two groups of patients. Clinicians should be mindful when comparing outcome MCID developed for a different population group in their practice.

In this study, we had expected patients to demonstrate significant improvement in wrist function measured with the PRWHE. The goal of the WSRP is to enable patients to return to performance of meaningful activities with their injured wrist. In the WSRP, there was a guided transition from isolated exercises at the wrist in early stages to exercises involving the entire upper extremity at later stages. The transition integrated the upper extremity into a kinetic chain where the extremity functions as a dynamic unit rather than as isolated segments. Individualized movement training specific to the patient's activities further trained the extremity for movement efficiency. The progression of exercises in the WSRP was similar to the DRUST program that comprised three stages.[20] In the DRUST program, the preparatory and strengthening phases focused on wrist movements while the last functional training phase emphasized the kinetic chain of the upper extremity.[20] The effects of the DRUST program were demonstrated with patients achieving clinical improvement in functions measured with PRWHE.[20] In this study, 70% of our patients demonstrated clinical improvement in PRWHE. There was also a large effect size of 2.47.

Researchers have advocated for rehabilitation that focused on movement patterns as they hypothesized that muscles in the upper extremity need to function synergistically, integrated in the kinetic chain of the extremity for performance of daily activities.[43] There are also studies that suggested that for patients with hand injuries, intervention focused on purposeful activities was superior to exercises which focused on ROM and strengthening in outcomes such as pain, ROM, functions and satisfaction.[44] Improvements in various aspects of body systems such as pain, grip strength, weight-bearing capability, and JPS could also have contributed to improvement in overall wrist function. Pain, strength and JPS have been shown to be significantly correlated with function.[4,41]

In our study, patients who were splinted for 4–6 weeks appeared to show greater clinical improvement in outcomes. These patients sustained a TFCC injury with associated DRUJ instability. The literature has recommended immobilizing patients with DRUJ

instability for 4–6 weeks in a stable position to reduce symptoms and allow healing prior commencing rehabilitation.[2] However, it is beyond the scope of this study to investigate the effects of splinting on recovery and patients' subsequent improvements in the outcome measures.

Feasibility of WSRP

WSRP has 4 stages with clear objective criteria for progression at each stage to assist therapists with clinical judgment. The physical examination and assessment of the wrist performed in the WSRP are fundamental skills which all hand therapists of various levels should be competent in. As WSRP is a pain-guided program, patients also played an active role in determining the level and intensity of the exercises by reporting if pain on NPRS exceed 2 points during exercise. Patient pain reporting would further confirm therapists' clinical judgment.

The flexibility in WSRP allowed for training specificity, which had been advocated to improve strength and functional performance of an individual.[21] As such, the total duration of WSRP varied for each patient depending on patients' symptoms and goals. In this study, patients took a median of 109 days to complete WSRP. Other wrist ligament rehabilitation programs described in the literature also lasted between three to six months.[19,20,45] This duration concurs with the ligament healing timeline reported in the literature. Ligament healing has been reported to require at least 60 days, up to 360 days for remodeling and maturation of the ligament to occur.[46]

In the WSRP, the frequency of follow up varied for each patient and was agreed upon between the therapist and the patient. As all patients were working adults who still attended work despite their wrist injury, the flexibility of the program allowed patients to plan their commitments and pace their rehabilitation progress. This flexibility also placed emphasis on the patient's responsibility to follow through the prescribed home exercises. In this study, the treating therapists noted down the home exercises or took photos of patients performing the exercises which served as visual cues for home exercises. With hindsight, an exercise journal with exercise pictures and exercise regimes could be provided for clarity.

In designing home exercises, there was careful consideration to ensure that patients could perform the exercises with commercial readily available exercise equipment such as dumbbells, a weighted ball, and theraband, to increase exercise adherence. The Baltimore Therapeutic Equipment, while only available in clinic, could simulate patient-identified activities to further train movement efficiency in specified activities.[28]

In this study, exercise compliance was not measured formally. As all patients had work commitments, patients reported difficulty complying to the daily exercise regime. Patients reported performing the home exercises at least three times a week. The positive result in grip strength suggests that the exercise dosage was adequate to improve strength. Reiman and Lorenz also recommended strengthening frequency of 3–4 days a week in rehabilitation program so that there are at least 24 hours between sessions for muscles to rest.[21] The rest ensured that there was optimal recovery between each training sessions to minimize risk of overtraining and subsequent injury.[47]

Limitations

There are limitations to this study. First, our patients did not undergo wrist arthroscopy to diagnose their TFCC injuries. Wrist arthroscopy is considered the gold standard in diagnosing TFCC injuries.[48] In our study, the specific type of TFCC injury could not be accurately determined and we were unable to analyze our patients' recovery based on the type of their TFCC injury. Next, the absence of a comparative group did not allow evaluation of WSRP true treatment effects compared to routine strengthening in therapy. Improvement of symptoms with rest and routine strengthening cannot be dismissed. However, the promising results from this study support further research with a typical treatment control group. Another limitation of this study was the flexible treatment duration which could affect conduct of future randomized controlled trials (RCTs). As most of our patients completed WSRP within 3 months, we suggest standardizing the treatment duration to 3 months in future RCTs. This concurs with Bonhof-Jansen et al.[20] who found in their TFCC study that their patients attained maximum effect of rehabilitation within 3 months. Next, results of this study cannot be generalized due to nature of the study and the small sample size. Lastly, there was no reliable method to assess JPS in a clinical setting. Studies have reported poor test-retest reliability of measurement of joint position sense with goniometry.[49] It is imperative that clinicians interpret the results from this study with caution and use astute clinical reasoning and judgment when applying WSRP concepts on patients. For patients with specific physical needs such as in athletes or heavy manual workers, they may require further sports rehabilitation or work hardening after completing the WSRP.

Conclusion

A WSRP was developed from careful synthesis of concepts described in the literature, and the author's clinical experience. The use of DTM and PNF in WSRP are novel approaches in TFCC wrist sensorimotor rehabilitation. This study provides a potential treatment approach in rehabilitation of TFCC injuries. Promising clinical improvements in wrist function, grip strength, pain, and weight bearing achieved by the patients in this study suggest the feasibility of WSRP. There is a need for larger prospective clinical trials with a control group to investigate the true treatment effects of proposed WSRP. Findings from this study could contribute to the evidence base of TFCC wrist sensorimotor rehabilitation.

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Declaration of Conflicting Interests

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Ethical approval

Ethical approval for this study was obtained from Singhealth Institutional Review Board (reference number: 2017/3149).

Informed consent

Written informed consent was obtained from all subjects before the study.

Guarantor

ZC.

Contributorship

Not applicable.

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